Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**
Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Programming Languages: Python, Java, C++, C, JavaScript, SQL
Frontend Development: React.js, HTML5, CSS3, Responsive Web Design
Backend Development: FastAPI, Flask, Node.js, REST APIs, API Development, Middleware, Microservices
Databases & Caching: MySQL, SQLite, MongoDB, Redis, SQLAlchemy
Software Engineering: DSA, OOPs, Multithreading, TCP Sockets, Design Patterns, Debugging, Unit Testing
AI/ML & Frameworks: Scikit-learn, TensorFlow, PyTorch, Generative AI, Retrieval-Augmented Generation (RAG), EasyOCR
Tools & Platforms: Git, GitHub, VS Code, Postman, Docker, Power BI
Cloud & Deployment: Render, Vercel, Google OAuth, CI/CD Fundamentals

EXPERIENCE

AI for Sustainability Virtual Intern* |LINK** — ***July – Sep, 2026

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built an OCR-based data extraction pipeline using EasyOCR, OpenCV, and NLP to transform unstructured receipts into structured transaction data, improving processing efficiency by 40%.
- Developed data preprocessing and expense categorization workflows to clean, validate, and organize extracted transaction data for downstream analysis.
- Built analytics and reporting features for spend trends and budget tracking, converting transaction data into actionable financial insights for end users.

***Docwise* |LINK**

- Built an AI-powered PDF question-answering system using Google Gemini and RAG, enabling contextual question answering over documents exceeding 200 pages.
- Developed a document processing and retrieval pipeline combining PDF extraction, chunking, embeddings, and semantic retrieval, achieving 95%+ query accuracy.
- Integrated FastAPI and LLM inference to deliver an end-to-end AI application for document understanding and conversational querying.

***RedisLite* |LINK**

- Engineered a Redis-inspired in-memory key-value database in Python featuring a custom RESP protocol parser, multithreaded TCP server, thread-safe storage engine, and an O(1) LRU cache with lazy/active TTL expiration.
- Designed a four-layer architecture (TCP transport, storage engine, cache/TTL, persistence) implementing Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery.
- Validated the system with 46 automated PyTest unit tests (TDD), benchmarking 619K SET ops/sec in-memory and 2.8K ops/sec with durable WAL persistence.

***API Rate Limiter Library* |LINK**

- Built a Python rate-limiting library implementing five throttling algorithms, Redis integration, and reusable Flask/FastAPI middleware for API automation and reliable traffic control.
- Designed a modular, thread-safe architecture with pluggable storage backends to support concurrent request handling and reliable API behavior across application environments.
- Developed 30+ PyTest automated tests covering concurrency, edge cases, and algorithm correctness, and packaged the library on PyPI for reusable software development.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** — 2026